

- 8 A ball rolls down a frictionless circular ramp as shown below.



Which statement is incorrect?

- A The force that the ramp exerts on the ball is always perpendicular to the velocity of the ball.
- B The force that the ramp exerts on the ball is greater than its weight when it is at the bottom of the ramp.
- C The speed of the ball when it goes around the circular ramp is not constant.
- D The acceleration of the ball is always directed towards the centre of the circular ramp.

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9 The gravitational potential energy of an object of mass  $2.0\text{ kg}$  at the surface of the Earth